

JESSE DONWAHOUE

Master's Application in Robotics

Embedded Systems & IoT

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LANGUAGES

- French: native
- English: technical (datasheets, docs, tutorials)

STRENGTHS

- Fast learner
- Analytical mindset
- Rigor & method
- Technical curiosity
- Adaptability
- Works well under pressure

INTERESTS

- Robotics & IoT
- Autonomous systems
- Bioinformatics
- Sports | Music
- Travel

CURRENTLY LEARNING

Protocols

I2C, SPI, UART

Control

PWM, PID (basics), Control Theory (coursework)

Actuators

Stepper motors, Dynamixel

PROFILE

Embedded electronics and automated systems technician, specialized in autonomous robotics and IoT, with a Bachelor's degree in Software Engineering as a complementary background.

I design autonomous robotic systems: sensors → embedded processing (ESP32/Arduino) → actuators.

Fast learner, technically rigorous, able to carry projects through from A to Z.

Master's project: deepen my expertise in mobile robotics and IoT (perception, control, ROS2) through a Master's program, in order to grow into embedded systems and autonomous robotics engineering.

EDUCATION

Degree / Program	Institution	Period
Embedded Electronics & IoT Training (ongoing)	ADESTI • Cotonou	Feb. 2026 – July 2026
Bioinformatics Training (ongoing)	CEDORIASE ACADEMY	April 2026 – present
Bachelor's Degree in Software Engineering	IFRI / UAC • Calavi	2021 – 2025
High School Diploma (Baccalauréat)	Complexe Scolaire Planète Éducative	2020 – 2021

CERTIFICATIONS

Technical Certifications

Certification	Organization	Date
Complete Arduino Course (5h31)	Cursa	July 2026
AI for Aerospace & Mechanical Engineers	Free Academy	June 2026
IoT (Internet of Things) – Wireless & Cloud Computing Emerging Technologies	Yonsei University (Coursera)	June 2026
Code Yourself! An Introduction to Programming	Univ. of Edinburgh & Univ. ORT Uruguay (Coursera)	June 2026

Additional Certifications

Certification	Organization	Date
AI Fundamentals	MTN Skills Academy / Microsoft Learn	May 2026
AI for Everyone	MTN Skills Academy / Microsoft Learn	May 2026
Generative AI	MTN Skills Academy / Microsoft Learn	May 2026
Content Marketing Certified	HubSpot Academy	May 2026 – June 2028

ROBOTICS & EMBEDDED PROJECTS

Autonomous Scarecrow Robot — 2 evolving versions

- ▶ Board: Arduino / ESP32
- ▶ PIR sensor: motion detection up to 7 m
- ▶ Actuators:
 - Version 1 (2026): controlled via electronic relays
 - Version 2 (2026): controlled via electrical contactors (more robust, higher loads)
- ▶ Embedded logic: interrupts + timers, automatic triggering of light / siren / servo

- ▶ Power supply: 12V battery, optimized standby power consumption
- ▶ Language: C++ (Arduino IDE)

This project demonstrates my ability to design an autonomous, scalable, and robust system under real-world conditions.

Electrical Consumption Monitoring System (IoT)

- ▶ Board: ESP32 (built-in Wi-Fi)
- ▶ Sensors: voltage and current (real-time measurement)
- ▶ Pipeline: sensors → ESP32 → embedded web interface (HTML/CSS/JS)
- ▶ Automatic email alerts via SMTP in case of overconsumption

Smart Parking System

- ▶ Board: Arduino
- ▶ Sensors: HC-SR04 ultrasonic (presence detection per spot)
- ▶ Outputs: LED display + signage panel
- ▶ Logic: automatic real-time management of available spots

TECHNICAL SKILLS — ROBOTICS & EMBEDDED SYSTEMS

Microcontrollers ESP32, Arduino (C/C++)
Sensors PIR, HC-SR04 ultrasonic, current/voltage, temperature, LDR
Actuators Relays, contactors, LEDs, buzzer, servo motor
Embedded Interrupts, timers, standby management, C++
Communication Wi-Fi (ESP32), I2C, SPI, UART
Software Robotics ROS (currently training, The Construct)
Related Web Dev HTML/CSS/JS, Vue.js, Node.js, REST API, SQL
Tools Git, VS Code, Figma, Arduino IoT Cloud

PROFESSIONAL EXPERIENCE		
Position	Company	Period
Mobile Developer	ACT • Abomey-Calavi	March – July 2024
Web Developer (internship)	Gits • Cotonou	Aug – Sept. 2023
Customer Service Agent	Carrefour / Concentrix • Cotonou	July 2025 – Jan. 2026

- ▶ ACT: Designed collection-tracking interface, integrated IoT sensors (light, temperature), database modeling
- ▶ Gits: Developed REST API with Node.js/Express.js, dynamic web interfaces
- ▶ Concentrix: Managed calls/emails/messages, effective communication under pressure